

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Justification

DrBenchmark is exclusively text-based French biomedical prose [Q31, Q48], with all data using the standard Latin alphabet and French diacritics. The deployment is also text-only metropolitan French regulatory documents — there is no script, modality, or infrastructure mismatch. Tokenization behavior is documented and benchmarked across models [Q78], with sub-token segmentation analyzed for impact on performance [Q79]. The benchmark uses HuggingFace Datasets/Transformers/PyTorch [Q48], directly compatible with the deployment's likely model integration stack. Minor concerns: PxCorpus contains spoken-dictation transcripts with ASR noise (PXCORPUS-D13, PXCORPUS-D15) and CAS includes patient-reported informal speech (CAS-D20, CAS-D21), creating register-level form mismatches with formal regulatory documents; some encoding artifacts observed (E3C-D15, E3C-D16, QUAERO-D15).

Strengths

- Exclusively text-only French in standard Latin alphabet — no cross-modality or script mismatch (universal across datasets, e.g., CAS-D1, CLISTER-D8, ESSAI-D11)
- HuggingFace toolkit with normalized data loaders matches likely deployment stack [Q48]
- Tokenization behavior empirically analyzed across 8 models with sub-token segmentation metrics reported [Q78, Q79]
- Metropolitan French infrastructure assumptions consistent with deployment context

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions match: both benchmark and deployment use written French text in Latin alphabet with French diacritics. No image/audio modality concerns. Tokenization analysis [Q78] provides model-level form characterization.

IF-2: Regional infrastructure (Metropolitan France, high-resource digital text) supports the same text-capture specifications. No infrastructure mismatch.

IF-3: Domain-specific form considerations: the deployment processes formal written regulatory templates (SmPC/PIL/CTD), which is matched by QUAERO EMEA and DEFT2020 leaflet content. PxCorpus introduces a form mismatch (spoken transcripts with ASR noise PXCORPUS-D13, PXCORPUS-D15, PXCORPUS-D17) that does not represent regulatory document form. Encoding artifacts (E3C-D15 backtick replacing accent; E3C-D16 escaped newlines) are minor but present.

ID	Evidence
PXCORPUS-D 13	i'll agree come on say avec successes merde je vais roulé faut lui faire un et mettre la rame en mode français — Code-switching and expletives from genuine recording transcription
PXCORPUS-D 15	/chet — ASR noise token, not a biomedical term

E3C-D15	bilirubine totale a ` 140 mmol/L — Backtick artifact replacing accent; minor tokenization quality concern
----------------	---

IF-4: Form mismatches are minor: (a) PxCorpus spoken transcript form is not representative of regulatory written documents; (b) document-splitting artifacts in QUAERO (QUAERO-D15, QUAERO-D17) produce vacuous fragments; (c) tabular/abbreviated data in CLISTER (CLISTER-D19, CLISTER-D20) and CAS (CAS-D16) atypical of regulatory prose. None of these is a fundamental form violation.

ID	Evidence
QUAERO-D15	o . — Single-character fragment artifact from document splitting
PXCORPUS-D17	ne pas tenir compte à midi tous les jours merd/ — Expletive and truncated token in clinical transcription
CLISTER-D19	Jour 45 - - - - - 25 µg 0,4 mg" / "Jour 53 15,13 12,0 1,58 - - - - - 25 µg 0,4 mg — Table rows with dash placeholders; structurally unlike regulatory prose

Information Gaps

- Sub-token segmentation impact specifically on long INN strings and EMA-template phrases not characterized